



ST BENEDICT'S

SUBJECT	Physical Science Paper 2	DATE	23 November 2020
GRADE	11	MARKS	200
EXAMINER	Mr du Toit	MODERATOR	Mr McCoy
NAME		DURATION	3 hours
CLASS			

LOW ORDER

55%

HIGH ORDER

45%

INSTRUCTIONS

1. This question paper consists of 11 pages, an Answer Sheet of 2 pages and a Data Sheet of 2 pages. Please make sure that your question paper is complete.
2. Answer ALL the questions.
3. Read the questions carefully.
4. Use the data and formulae whenever necessary.
5. **Start each question on a new page.**
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in all answers.
8. Express all answers to TWO decimal places unless told otherwise.
9. It is in your own interest to write legibly and to present your work neatly.

Question	1	2	3	4	5	6	7	8	9	TOTAL
Total	20	21	20	30	29	28	13	17	22	200

QUESTION 1 – MULTIPLE CHOICE**20 MARKS**

Answer the following multiple-choice questions on your separate ANSWER SHEET. Make a cross (X) in the box corresponding to the letter which you consider to be the most correct.

1.1 The chemical formula for aluminium hydrogen sulphate is:

- A $\text{Al}(\text{HSO}_4)_3$
- B $\text{Al}_3\text{H}_2\text{SO}_4$
- C $\text{Al}_2(\text{HSO}_4)_3$
- D $\text{Al}_2\text{H}_2(\text{SO}_4)_3$

(2)

1.2 Which sequence of Group 18 elements demonstrates a gradual decrease in their boiling points? All the choices are elements in the liquid state.

- A Ar, Kr, Ne, Xe
- B Ne, Ar, Kr, Xe
- C Xe, Kr, Ar, Ne
- D Kr, Xe, Ar, Ne

(2)

1.3 In a series of experiments, 0,05 g samples of magnesium were added separately to 100 cm³ volumes of hydrochloric acid. The table below summarises the experimental conditions:

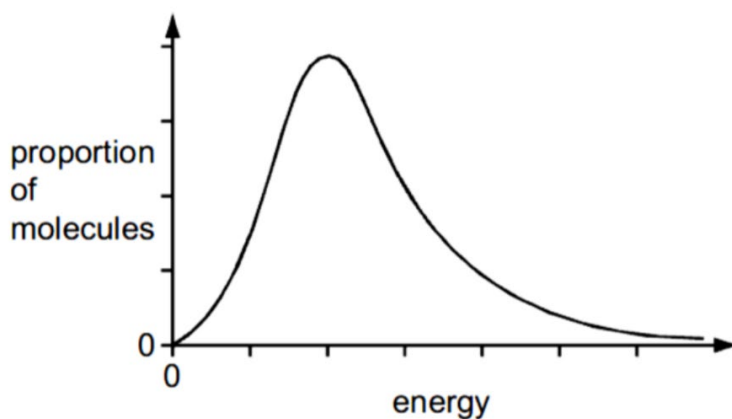
Experiment	Mg(s)	Concentration of HCl(aq) (mol·dm ⁻³)	Temperature of acid (°C)
I	Ribbon	0,1	25
II	Ribbon	0,1	20
III	Powder	0,1	70
IV	Powder	0,5	25
V	Powder	0,1	20

In which of the experiments would the magnesium be expected to take the shortest times to react completely with the excess hydrochloric acid?

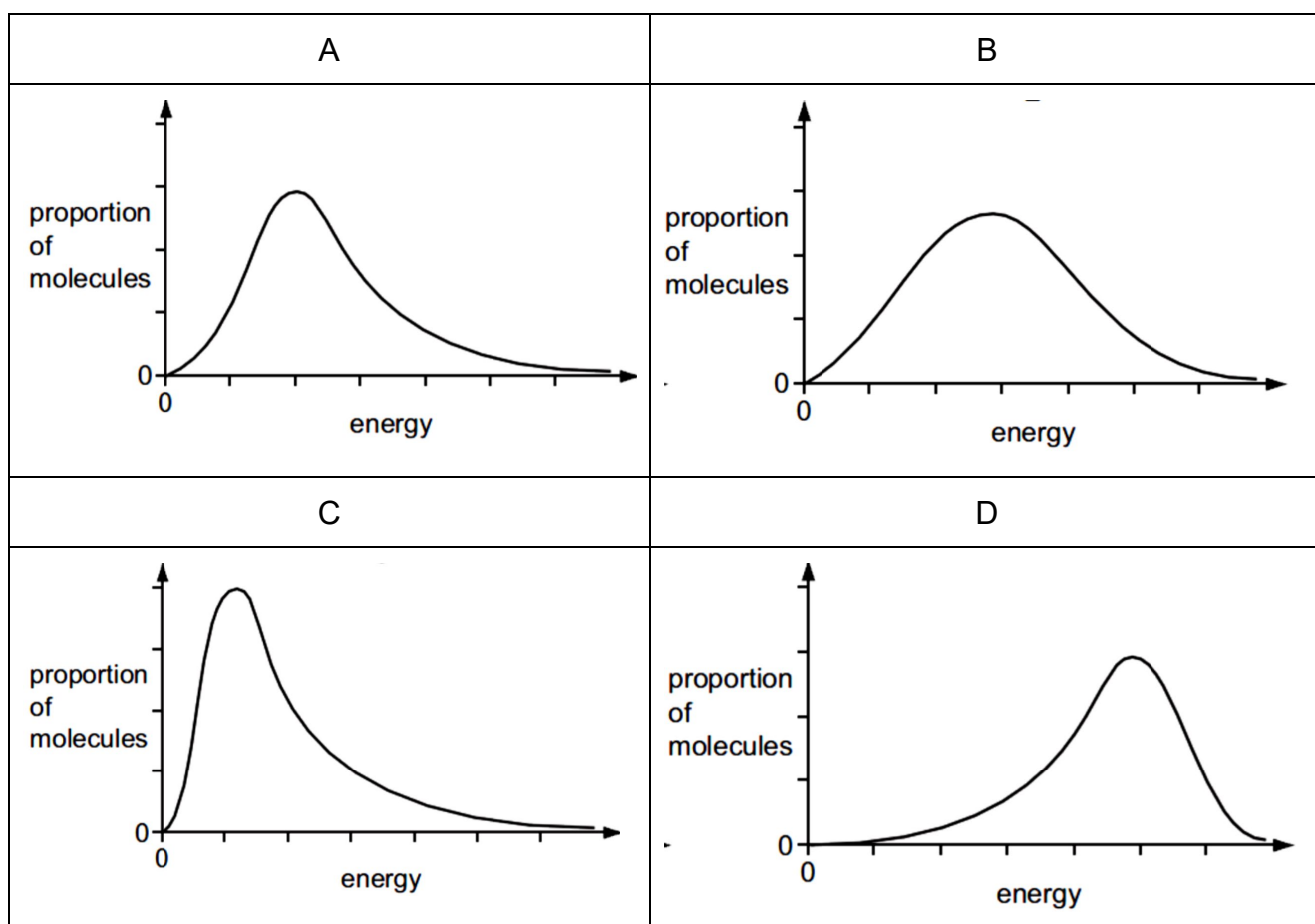
- A I and II
- B III and IV
- C III and V
- D II and IV

(2)

- 1.4 The Maxwell-Boltzmann energy distribution curve represents the variation in energy of the molecules of a gas at room temperature.



Which curve applies for the same gas at a lower temperature?



(2)

- 1.5 Which of the following represents one mole?

- A 22,4 g of liquid water
- B 22,4 dm³ of liquid water
- C 18 g of liquid water
- D 18 dm³ of liquid water

(2)

1.6 How many **atoms** are present in 2 mol of hydrated oxalic acid, $(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$?

- A $1,20 \times 10^{24}$
- B $1,69 \times 10^{25}$
- C $3,37 \times 10^{25}$
- D $5,78 \times 10^{25}$

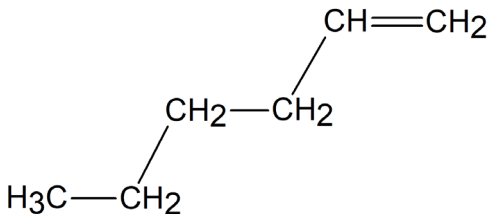
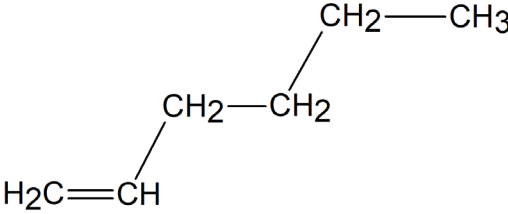
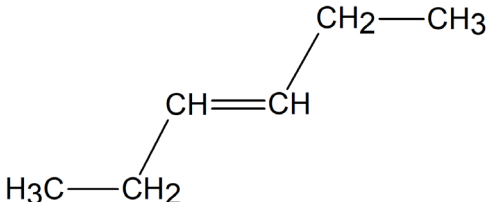
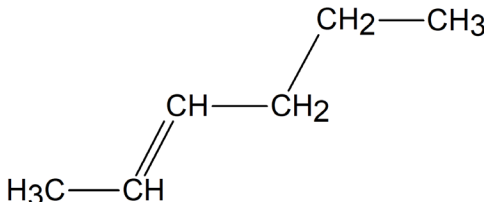
(2)

1.7 Which of the following compounds is an alkane?

- A ethene
- B 2-bromopentane
- C 2-methyl-3,4-dichloropentane
- D 2,3-dimethyl-4-ethyloctane

(2)

1.8 Hexan-2-ol undergoes dehydration in the presence of concentrated H_2SO_4 . Which of the following CANNOT be a product of this dehydration?

A	B
	
C	D
	

(2)

1.9 Which of the following haloalkanes CANNOT undergo a dehydrohalogenation reaction?

- A 2-bromobutane
- B 2,2-dichloropropane
- C trichloromethane
- D 1,2-dibromobutane

(2)

1.10 2-methylbutane and pentane are molecules that illustrate the following types of isomerism when compared with each other:

- A chain isomerism
- B positional isomerism
- C functional group isomerism
- D stereoisomerism

(2)

QUESTION 2

21 MARKS

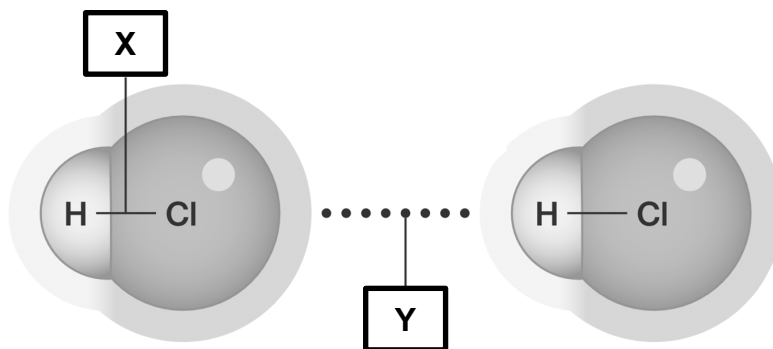
2.1 Consider the molecules below:

Letter	Formula or name
A	K_2SO_4
B	KI
C	SiO_2
D	Magnesium
E	NH_3
F	Water
G	$HCOOCH_3$

Answer Question 2.1.1 – 2.1.5 on the ANSWER SHEET provided. Answers may have more than one correct answer. Write all the correct answers to a question. Incorrect answers will be penalised. Write down only the LETTER/S that correctly correspond.

- 2.1.1 exhibits hydrogen bonding between molecules
- 2.1.2 conducts electricity in the molten state
- 2.1.3 a polar molecule
- 2.1.4 consists of positive kernels and a sea of delocalised electrons
- 2.1.5 a giant covalent structure that exhibits a high melting point (10)

2.2 Consider the diagram below, showing the arrangement of two hydrogen chloride molecules in liquid phase:



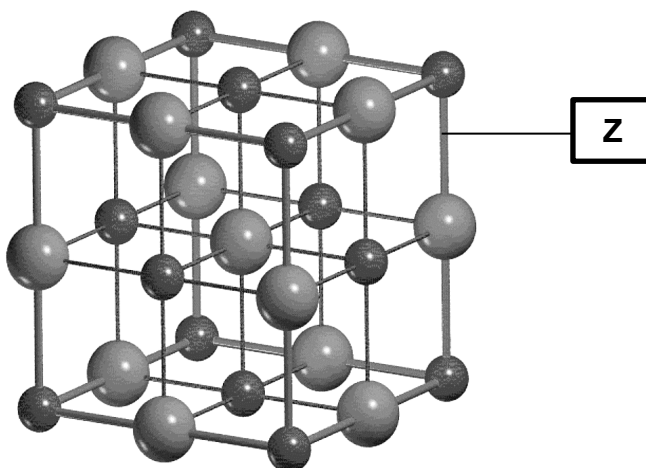
- 2.2.1 Define the term *intramolecular bond*. (2)
- 2.2.2 Name the specific type of intramolecular bond represented by the letter **X** in the diagram. (2)
- 2.2.3 Fully explain your answer in 2.2.2 by referring to your understanding of electronegativity. (3)
- 2.2.4 Define the term *intermolecular force*. (2)
- 2.2.5 Name the specific type of intermolecular force represented by the letter **Y** in the diagram. (1)
- 2.2.6 What is the partial charge (δ^+ or δ^-) on the hydrogen atom in the HCl molecule? (1)

QUESTION 3

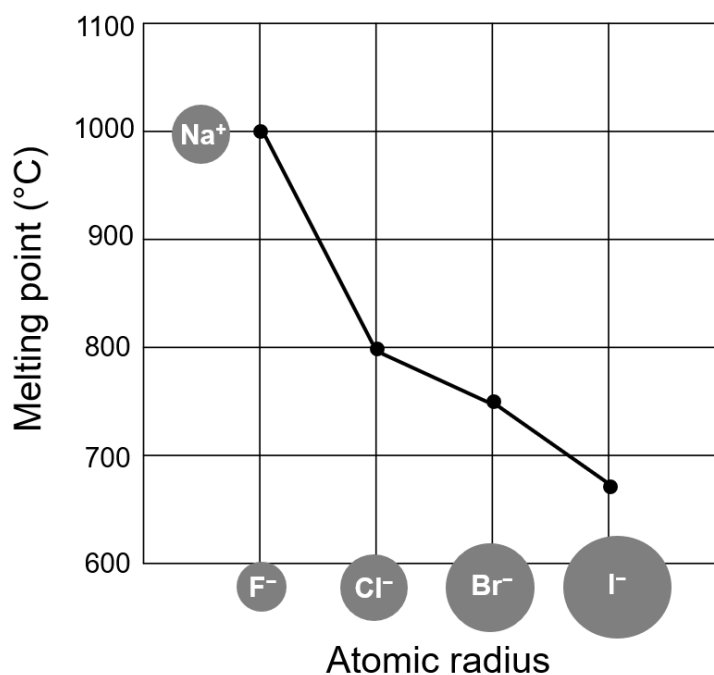
20 MARKS

Sodium fluoride is formed from oxidation and reduction half reactions.

- 3.1 Define *oxidation*. (2)
- 3.2 Write the oxidation half reaction of the sodium atom. (2)
- 3.3 Define *oxidising agent*. (2)
- 3.4 Which atom would be the oxidising agent? State either SODIUM or FLUORINE. (1)
- 3.5 The diagram below shows the crystal lattice structure of the giant ionic solid sodium fluoride.



- 3.5.1 Under which conditions is the above substance able to conduct electricity? Explain your answer. (3)
- 3.5.2 Provide a label for **Z**. (1)
- 3.5.3 With reference to the crystal lattice structure, explain why sodium fluoride has a very high melting point (1000°C). (3)
- 3.6 The following graph shows the melting points of different sodium halides.



- 3.6.1 Describe the trend in the above graph. (2)
- 3.6.2 Use your knowledge of electrostatics to explain the trend described in 3.6.1. (4)

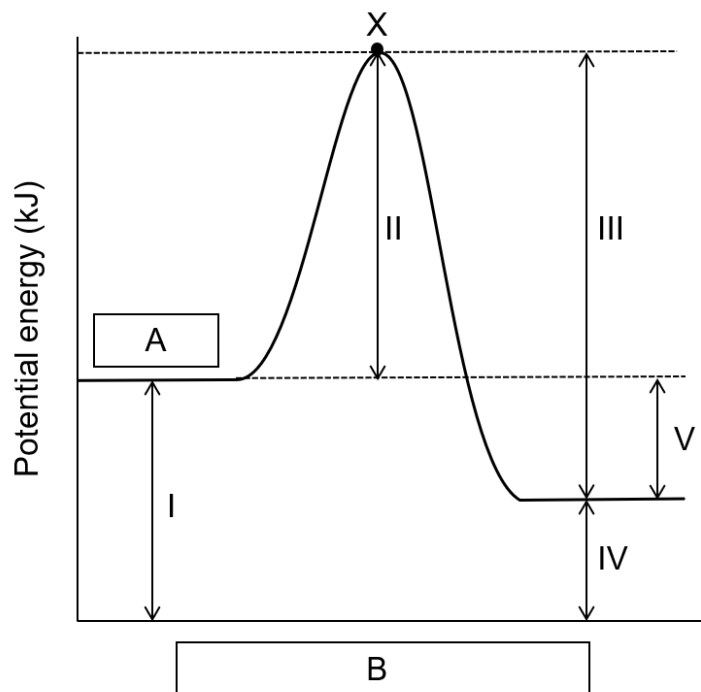
QUESTION 4

30 MARKS

A sample of calcium carbonate and hydrochloric acid react in a reaction vessel. The balanced chemical equation for this reaction is:



Consider the potential energy profile for this reaction as illustrated below. The graph has been labelled I – IV to represent the various energies that are illustrated by this energy profile. The graph has been reproduced for you on your ANSWER SHEET.

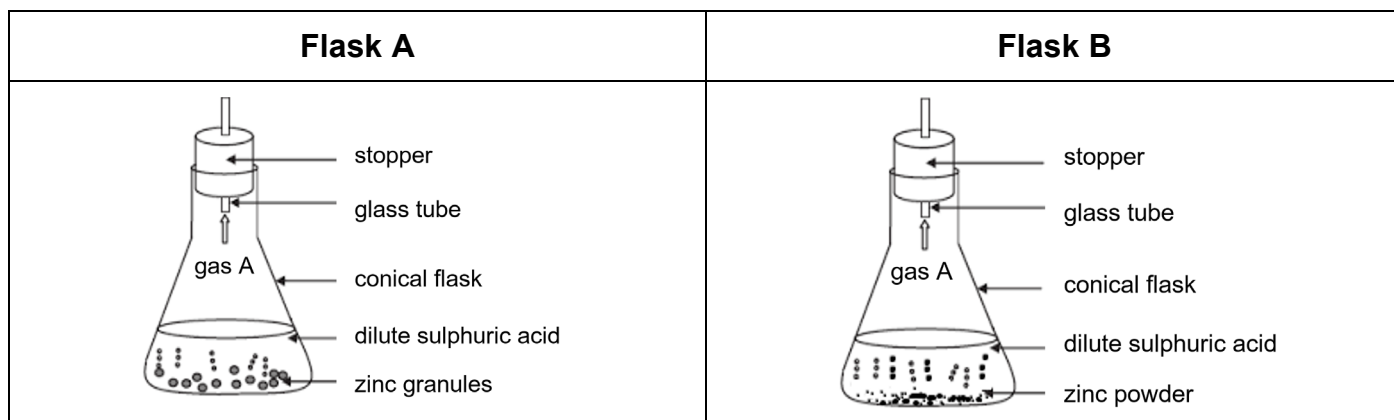


- 4.1 Define the energy labelled V. (2)
- 4.2 Complete the following on your ANSWER SHEET:
- 4.2.1 Provide the names of the energies (I to IV) in the spaces provided. (4)
- 4.2.2 Provide the labels for A and B in the space provided. (2)
- 4.3 Consider point X.
- 4.3.1 Provide a name for the position on the graph labelled X. (1)
- 4.3.2 State two conditions of the reacting particles that must be met in order for X to form. (3)
- 4.3.3 What two processes are occurring simultaneously at X? (2)
- 4.4 Name the type of reaction this graph represents. Explain how you came to this conclusion. (3)
- 4.5 The same reaction takes place in the presence of a catalyst. On the potential energy profile on the ANSWER SHEET, show how the graph would change in the presence of a catalyst using a dotted line. (2)
- 4.6 How would the presence of a catalyst affect the value of the energy labelled V? Explain your answer. (3)
- 4.7 A 3 g sample of calcium carbonate and hydrochloric acid react in a reaction vessel. 0,5 dm³ of carbon dioxide gas is collected at STP. Assume the reaction to be 100% efficient.
- 4.7.1 Calculate the number of moles of carbon dioxide formed in the reaction at STP. (3)
- 4.7.2 Hence, calculate the percentage purity of the sample. (5)

QUESTION 5

29 MARKS

A student wants to investigate how the surface area of reactants influences the rate of reaction between zinc and excess dilute sulphuric acid. He sets up two conical flasks as shown in the diagram below. He records the volume of gas A collected using equal amounts of zinc.



5.1 Define *reaction rate*. (2)

5.2 Write a balanced chemical equation for the reaction between zinc and sulphuric acid. (3)

5.3 For this investigation, identify:

5.3.1 the independent variable (1)

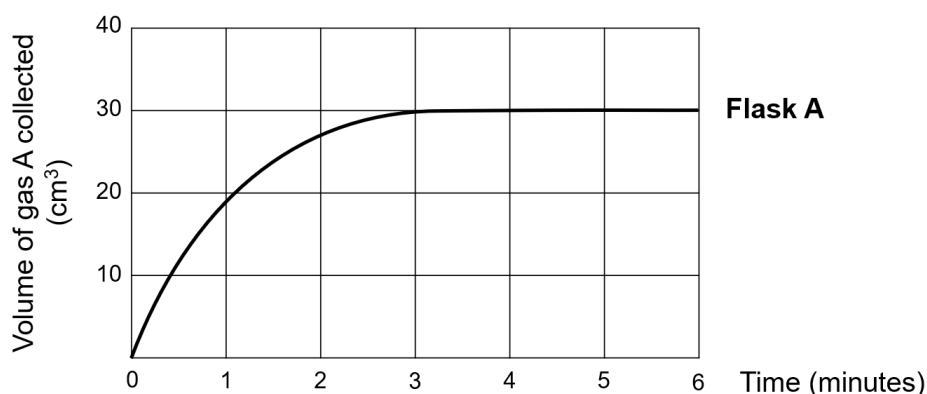
5.3.2 the dependent variable (1)

5.3.3 two controlled variables (2)

5.4 Explain why the variables you mentioned in 5.3.3 have to be controlled. (2)

5.5 Another student states that the rate of this reaction can also be suitably measured by placing the reaction vessel on a mass scale and measuring the decrease in mass over time. Do you agree with this student? Briefly explain. (3)

The following graph represents the students results.



5.6 Calculate the average rate of the reaction in flask A between 0 and 3 minutes. (3)

5.7 Explain why the gradient of the graph decreases over time. (2)

5.8 The above graph has been reproduced for you on your ANSWER SHEET. On this graph, sketch the graph that represents the results from Flask B. (3)

5.9 Explain your answer in 5.8 by referring to the collision theory. (4)

5.10 A student makes the following statement:

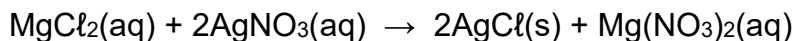
*When increasing the volume of solution (acid) in a 100% efficient reaction, the amount of product produced will **always** be greater as there will be moles of solute to react to form product.*

Evaluate this statement. (3)

QUESTION 6

28 MARKS

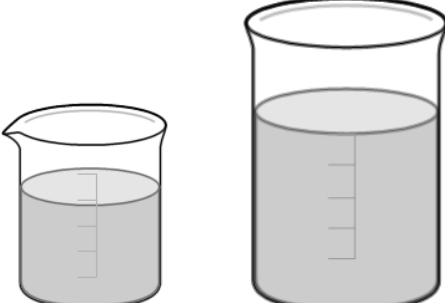
- 6.1 Two solutions, one being silver nitrate (AgNO_3) and the other, magnesium chloride (KCl) are mixed together in a glass container. Solid silver chloride (AgCl) forms at the bottom of the container as seen in the following balanced reaction that took place:



55 cm^3 of the magnesium chloride, with a concentration of $0,25 \text{ mol} \cdot \text{dm}^{-3}$ was mixed with 70 cm^3 of silver nitrate solution with a concentration of $0,15 \text{ mol} \cdot \text{dm}^{-3}$.

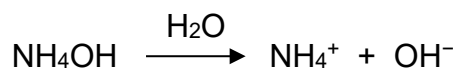
- 6.1.1 Prove that the silver nitrate is the limiting reactant. (6)
- 6.1.2 Hence, calculate the mass of silver chloride that can formed during this reaction. (3)
- 6.1.3 Calculate the percentage yield of silver chloride if 1,49 g of silver chloride was in fact recovered (3)

- 6.2 You are given the following two aqueous solutions:

Solution A $c = 0,2 \text{ mol} \cdot \text{dm}^{-3}$ ammonium hydroxide $V = 0,25 \text{ dm}^3$		Solution B $c = 0,3 \text{ mol} \cdot \text{dm}^{-3}$ ammonium carbonate $V = 0,5 \text{ dm}^3$
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- 6.2.1 Give a reason why the above solutions are *standard solutions*. (1)
- 6.2.2 Define the term *solvent*. (2)
- 6.2.3 Identify the solvent used in the above solutions. (1)
- 6.2.4 Give the method for making a standard solutions using four key bullet points. (4)

The balanced dissociation reaction for ammonium hydroxide is given as:

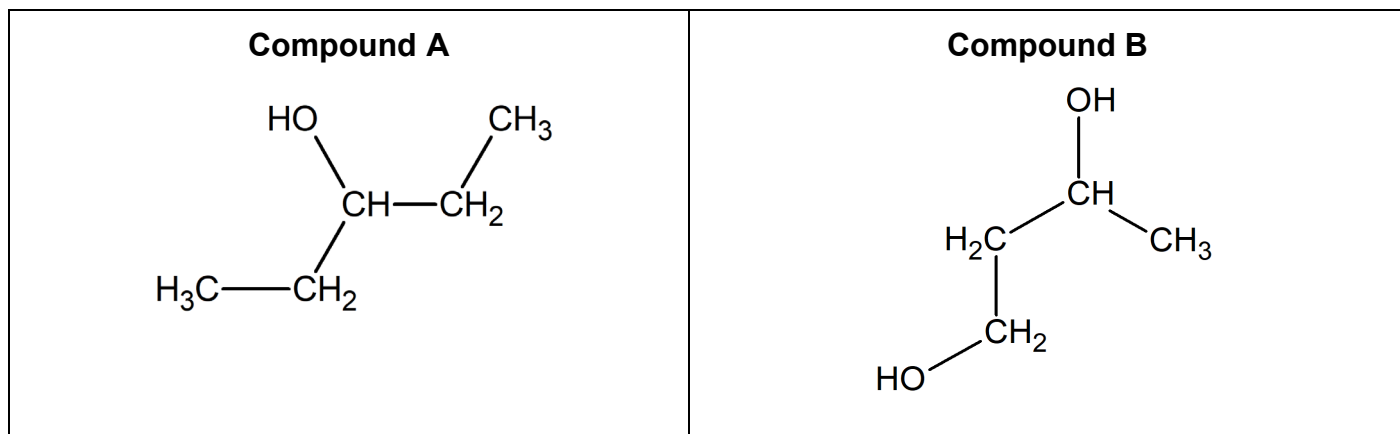


- 6.2.5 Write balanced chemical equation showing the dissociation of ammonium carbonate. (3)
- 6.2.6 Solution A is now mixed with Solution B in a new bigger beaker. Calculate the new concentration of the ammonium ions in the solution. (5)

QUESTION 7

13 MARKS

Consider the organic compounds below.

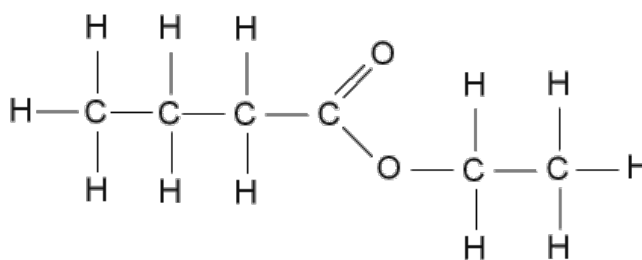


- 7.1 Define *functional group*. (2)
- 7.2 Using structural formula draw the functional group of the above compounds. (1)
- 7.3 Give the name of the functional group drawn in 7.2. (1)
- 7.4 Give the IUPAC name of compound B. (4)
- 7.5 Compound A and B have similar molecular masses.
- 7.5.1 Give the name of the predominant intermolecular force between the molecules of the above substances. (1)
- 7.5.2 Which compound (A or B) will have a higher boiling point? Fully explain your answer. (4)

QUESTION 8

17 MARKS

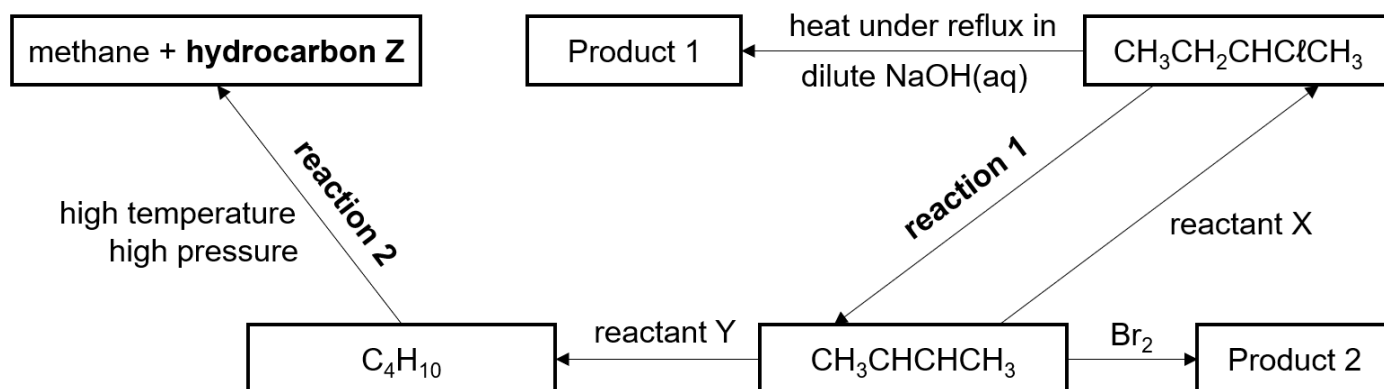
The structure below shows an organic compound that is used to give a pineapple flavour in sweets.



- 8.1 Write down the IUPAC name for the organic compound above. (2)
- 8.2 What homologous series does this compound belong to? (1)
- 8.3 Write down the IUPAC name of the carboxylic acid used to make this compound. (2)
- 8.4 Using structural formulae, draw the two reactants required to make this product. (4)
- 8.5 Write down the chemical formula of the acid catalyst used in the reaction that forms this compound. (2)
- 8.6 Define the term *structural isomers*. (2)
- 8.7 Write down the IUPAC name of one positional isomer of this compound. (2)
- 8.8 Write down the condensed structural formula of the functional isomer of this compound. (2)

QUESTION 9**22 MARKS**

Refer to the flow diagram below and answer the questions that follow:



- 9.1 Explain why $\text{CH}_3\text{CHCHCH}_3$ is an unsaturated hydrocarbon. (2)
- 9.2 Give the GENERAL NAME for:
- 9.2.1 Reaction 1. (1)
- 9.2.2 Reaction 2. (1)
- 9.3 Draw the structural formula for Product 1. (2)
- 9.4 Give the reaction conditions for Reaction 1. (2)
- 9.5 Give the chemical formula of:
- 9.5.1 Reactant X. (1)
- 9.5.2 Reactant Y. (1)
- 9.6 Write a balanced chemical equation for the complete combustion reaction of $\text{CH}_3\text{CHCHCH}_3$ using molecular formula. (3)
- 9.7 $\text{CH}_3\text{CHCHCH}_3$ is reacted with bromine (Br_2) in order to form Product 2. The gas is bubbled through red-brown bromine water.
- 9.7.1 What is observed when the gas is bubbled through the bromine water? (2)
- 9.7.2 What SPECIFIC TYPE of organic reaction is this? (1)
- 9.7.3 Draw the structural formula of Product 2. (2)
- 9.7.4 Give the IUPAC name of Product 2. (2)
- 9.8 Reaction 2 is a cracking reaction.
- 9.8.1 What type of cracking is taking place in reaction 2. (1)
- 9.8.2 Give the general formula for hydrocarbon Z. (1)

END OF EXAMINATION